

Stellar Classification Using Machine Learning: A Comparative Analysis of Random Forest and Logistic Regression

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Abstract—In this project, we use photometric data from the Sloan Digital Sky Survey (SDSS) to classify celestial objects into three distinct categories: Stars, Galaxies, and Quasars. Through data visualization and Feature Engineering, we leverage physical properties such as color indices and redshift to separate the objects. We train and evaluate two machine learning algorithms: a Logistic Regression model and a Random Forest Classifier. The Random Forest model demonstrates superior performance, achieving an accuracy of 98.1%.

Index Terms—Stellar Classification, SDSS, Random Forest, Logistic Regression

I. INTRODUCTION

Modern astrophysics has entered an era of big data, driven by advanced observatories capable of scanning the night sky with unprecedented depth and resolution. The Sloan Digital Sky Survey (SDSS) is one of the leading astronomical surveys, having mapped millions of celestial objects. However, the enormous amount of data produced by the SDSS makes manual analysis simply impossible. To fully make use of this information, developing automated and highly accurate classification systems has become essential in computational astronomy.

The main goal of this project is to build machine learning models that can classify celestial objects into three categories: Stars, Galaxies, and Quasars (QSOs). To provide context for this classification task, it is essential to briefly define our target classes. **Stars** are luminous spheres of plasma located relatively close to Earth, primarily within our own Milky Way galaxy. **Galaxies** are massive, gravitationally bound systems consisting of billions of stars, gas, and dust, situated far beyond our local neighborhood. Finally, **Quasars** (Quasi-Stellar Objects or QSOs) are the extremely luminous active cores of distant galaxies powered by supermassive black holes (Carroll and Ostlie, 2017).

To achieve this goal, we used an SDSS dataset sourced from Kaggle (Kaggle, 2022), which contains 100,000 observations. Rather than working with images, we rely on a structured tabular format consisting of 17 features. Among these, we specifically focus on the photometric data from five different filters—ultraviolet, green, red, near-infrared, and infrared (u, g, r, i, z)—along with the redshift of each object.

Traditionally, classifying these objects required visually inspecting their shapes or taking detailed spectroscopic measurements. While spectroscopy is highly accurate, obtaining

the light spectrum for every object in the sky is expensive and highly time-consuming. Our approach, instead, uses photometric data, which is much faster and cheaper to collect, and feeds it into machine learning algorithms. By engineering features like physical color indices and training models such as Logistic Regression and Random Forest, we want to show that machine learning is a highly efficient alternative, capable of processing thousands of objects in seconds without compromising accuracy.

II. THE METHODOLOGY

A. System Overview

The architecture of this project follows a clear machine learning pipeline, designed to process raw astronomical data and output reliable celestial classifications. As illustrated in Figure 1, the workflow begins with the ingestion of the SDSS dataset. Once loaded, the data undergoes a cleaning and preparation phase. As a first step in data preprocessing, the dataset was inspected for missing or invalid values and duplicated rows. Ensuring data integrity is vital because machine learning models cannot process null values mathematically, erroneous data can distort the learned physical relationships, and duplicates can introduce bias during training.

After confirming the dataset’s integrity, we applied feature engineering techniques to extract meaningful physical properties, such as astronomical color indices, which help separate the different object classes. Following an in-depth exploratory data analysis, which included evaluating feature correlations, the pipeline diverges to create two specialized datasets tailored to our specific algorithms. One dataset was standardized and stripped of highly correlated variables to satisfy the strict mathematical requirements of Logistic Regression, while the other retained all engineered features in their original scale to leverage the robust, non-linear capabilities of the Random Forest classifier.

In the final stages, both prepared datasets were divided into training and testing sets. The models were subsequently trained on their respective data tracks and evaluated using various performance metrics to determine and compare their classification accuracy.

B. Exploratory Data Analysis

Before proceeding with model training, we performed an Exploratory Data Analysis to assess the quality of the data

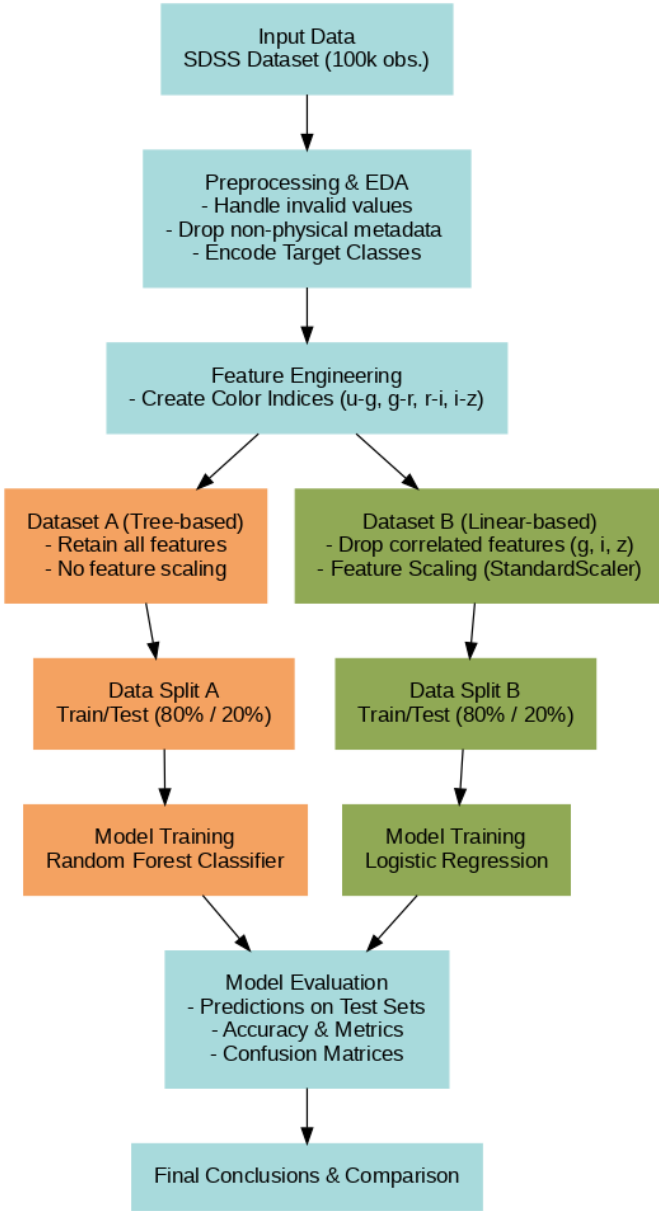


Fig. 1. Overview of the system's pipeline components and their interactions.

and understand the relationships between the features.

The initial inspection of numerical histograms revealed significant distortions in the distributions of the u , g , and z filters. This effect was caused by the presence of extreme values (specifically -9999), which function as placeholders for invalid measurements or missing data within the SDSS database. Identifying these entries was a crucial step, as their inclusion would have introduced noise and distorted the physical scales that the models need to learn.

Furthermore, we analyzed feature interactions using a correlation matrix and a pairplot analysis. The results, as illustrated in Figure 2, confirmed a high degree of multicollinearity among the five photometric filters (u , g , r , i , z), indicating that these variables provide highly redundant information. While

this redundancy is manageable for tree-based models, it can negatively impact the stability of linear models like Logistic Regression.

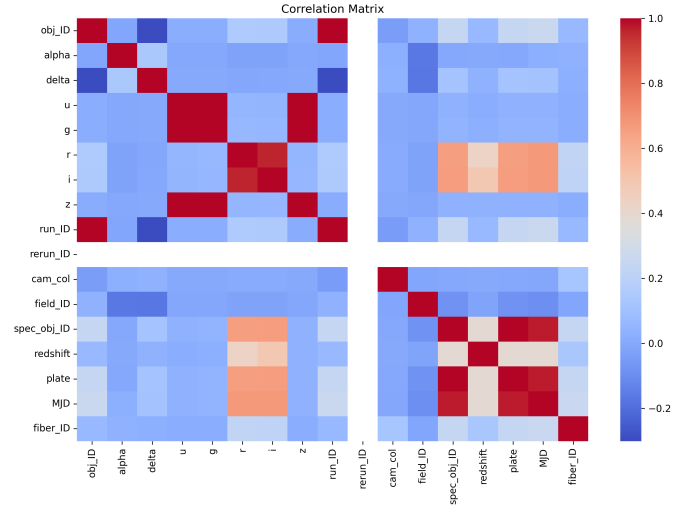


Fig. 2. Correlation matrix of the dataset features, highlighting strong multicollinearity among the photometric filters.

The visual analysis also provided early insights into class separability. As observed in the pairplot, *redshift* emerged as the most discriminative feature, showing a clear separation of Quasars (QSOs) from the other classes. In contrast, Stars and Galaxies exhibited a more significant overlap in their photometric profiles, suggesting that distinguishing between these two categories would require more complex decision boundaries. These findings directly informed our strategy to maintain two separate versions of the dataset, tailoring the data to the specific strengths and requirements of each algorithm.

C. Data Preprocessing and Feature Engineering

The initial dataset, sourced from Kaggle's SDSSDR17, consisted of 100,000 observations across 17 features. Guided by the insights from our Exploratory Data Analysis, we began the cleaning procedure by identifying the instrumental placeholder values (-9999), replacing them with null values (NaNs), and subsequently dropping these invalid entries. We also verified the absence of duplicate rows, resulting in a clean and robust dataset ready for feature extraction.

To ensure the models learned true astrophysical properties, we removed several metadata columns: *obj_ID*, *run_ID*, *rerun_ID*, *camcol*, *field_ID*, *spec_obj_ID*, *plate*, *MJD*, and *fiber_ID*. Theoretically, these features represent telescope operational data and observation IDs, which have no physical meaning regarding the celestial objects themselves. This logical decision was empirically reinforced by our histogram analysis, which demonstrated that some of these identifiers (such as *rerun_ID*) possessed zero variability across all 100,000 observations, thus offering no learning value. Retaining them could also lead to data leakage, where the model artificially achieves high accuracy by associating specific IDs with target

classes. Additionally, we chose to drop the celestial coordinates *alpha* (Right Ascension Angle) and *delta* (Declination Angle). While spatial distribution can be informative, relying on location creates a critical flaw for a general classifier. If the model learns to classify based on the specific regions mapped by the SDSS telescopes, it could fail to generalize to data from other observatories.

By dropping these coordinates, we force the model to classify objects based purely on their physical properties. Consequently, the core features retained for our analysis are the five photometric filters—ultraviolet (*u*), green (*g*), red (*r*), near-infrared (*i*), and infrared (*z*)—which measure the object’s brightness across different wavelengths, along with the object’s *redshift*, which provides crucial information regarding its relative velocity and distance from Earth.

To further enrich the dataset, we applied feature engineering guided by astronomical domain knowledge. In observational astronomy, an object’s “color” is mathematically defined as the difference in magnitudes between two photometric filters. These differences act as a direct indicator of an object’s surface temperature and spectral emission. Understanding this thermal profile is crucial for classification, as Stars, Galaxies, and Quasars possess distinctly different temperature ranges. To capture this physical reality, we computed four standard color indices: $u - g$, $g - r$, $r - i$, and $i - z$ (Abraham et al., 2011).

Despite creating these new features, we intentionally retained the original raw filter values (u, g, r, i, z) in our dataset. While the color indices provide information about temperature, the absolute magnitudes provide essential context regarding an object’s overall luminosity. As illustrated in Figure 3, plotting these engineered color differences reveals distinct clustering patterns, effectively separating Stars from Quasars and Galaxies based on their unique spectral distributions.

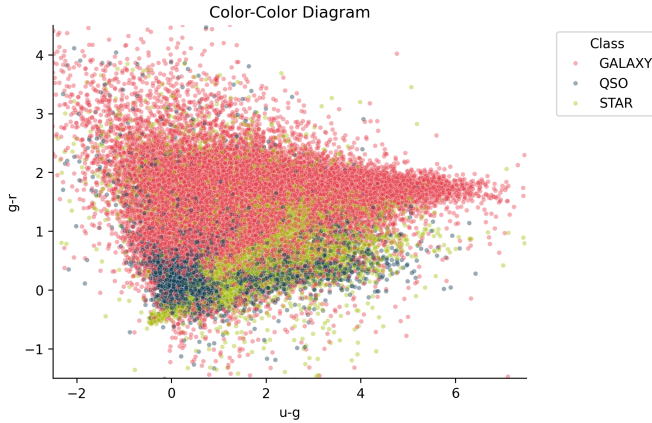


Fig. 3. Scatter plot of color indices illustrating the physical separation between celestial classes based on their temperature profiles.

Following the creation of these features, we addressed the multicollinearity identified during the Exploratory Data Analysis by branching our preprocessing pipeline into two distinct tracks:

- **Dataset A (Full Features):** Retained all original filters and engineered color indices. This comprehensive dataset was tailored for the Random Forest classifier, which is intrinsically robust to feature redundancy and can extract value from all available data.
- **Dataset B (Reduced Features):** To accommodate the Logistic Regression model, which is highly sensitive to multicollinearity, we dropped the highly correlated raw filters (g, i, z). This reduction stabilizes the linear model’s mathematical solver while preserving the necessary physical information through the remaining features.

Finally, we applied distinct scaling strategies tailored to the mathematical requirements of each model. The numerical features in Dataset B were standardized using *StandardScaler*. This step is strictly necessary for the Logistic Regression model, as it ensures that features with larger numerical ranges (like redshift) do not disproportionately dominate the learning process compared to features with smaller ranges (like the color indices). In contrast, Dataset A was intentionally left in its original scale. Because the Random Forest classifier relies on rule-based splits rather than distance calculations or gradient descent, it is immune to varying feature magnitudes. Keeping the data unscaled for this model not only is simpler but also preserves the physical interpretability of the features.

D. Machine Learning Models

In this phase of the pipeline, we implemented two distinct machine learning algorithms to classify the celestial objects: Logistic Regression and Random Forest. These models were selected to contrast a simple linear approach with a more robust non-linear ensemble method, with each algorithm utilizing the dataset specifically tailored to its mathematical requirements.

Logistic Regression was chosen as our baseline model and was trained on Dataset B. The primary goal of deploying this algorithm was to establish a foundational performance metric and to test whether the astronomical classes (Star, Galaxy, Quasar) could be effectively separated using linear decision boundaries. Because linear models are highly sensitive to both multicollinearity and the magnitude of input variables, the removal of correlated filters and the standard scaling applied to Dataset B were essential to ensure that the algorithm’s mathematical solver converged correctly and evaluated all remaining features equally.

To capture deeper patterns in the dataset, we trained a Random Forest classifier using Dataset A, which retained all original and engineered features in their unscaled form. This algorithm is an ensemble learning method that constructs a multitude of decision trees during training and aggregates their outputs to produce a more accurate and stable prediction. We selected Random Forest specifically for its powerful ability to model complex and non-linear relationships inherent in the astronomical data, which a linear model might miss. Furthermore, tree-based models are naturally robust to multicollinearity, different feature scales, and outliers, making them

exceptionally well-suited to for handling our diverse set of physical parameters.

E. Training Strategies and Hyperparameter Tuning

Each model was trained using its respective prepared dataset—Dataset A for the Random Forest and Dataset B for the Logistic Regression—to ensure an optimal learning process tailored to their specific mathematical requirements. To guarantee the reproducibility of our experiments, a fixed random seed (*random_state*=14) was established across all algorithms and data splitting procedures.

For the Random Forest classifier, the primary configured hyperparameter was the number of decision trees in the ensemble, which was set to 100 (*n_estimators*=100). This value provides an excellent balance: it is large enough to ensure stable and highly accurate predictions, but small enough to keep computational training times efficient. Aside from this, the model utilized the default hyperparameters provided by the *scikit-learn* library, allowing the trees to expand naturally without explicitly restricting their maximum depth.

Regarding the Logistic Regression baseline, the core adjustment was increasing the maximum number of iterations to 1000 (*max_iter*=1000). The default limit in *scikit-learn* (typically 100 iterations) is sometimes insufficient for the algorithm’s mathematical solver to fully converge and find the optimal decision boundary. Increasing this limit ensures that the training process completes successfully without early stopping errors.

III. EXPERIMENTAL SETUP

A. Dataset Split

To evaluate the predictive performance of our models, the final dataset of 100,000 observations was partitioned into two distinct subsets. This splitting procedure was applied identically to both Dataset A and Dataset B. We allocated 80% of the data (80,000 samples) to the training set, allowing the algorithms to learn patterns and physical relationships. The remaining 20% (20,000 samples) was reserved as a testing set to rigorously evaluate the models on unseen data. A fixed random seed (*random_state*=14) was utilized during this splitting process to ensure that the exact data distribution is preserved across executions, guaranteeing the reproducibility of the experiment.

Prior to training, it is crucial to analyze the class distribution within the dataset, as severe imbalances can bias a classifier toward the majority class. As illustrated in Figure 4, the dataset exhibits a natural imbalance that reflects real-world astronomical observational frequencies: Galaxies constitute the majority class, followed by Stars, while Quasars (QSOs) represent the minority.

Despite this uneven distribution, the absolute number of samples for the minority class remains significantly large (with thousands of observations). Because the models still have more than enough information to learn how to correctly identify them, we concluded that applying methods to artificially balance the classes was not necessary for this project.

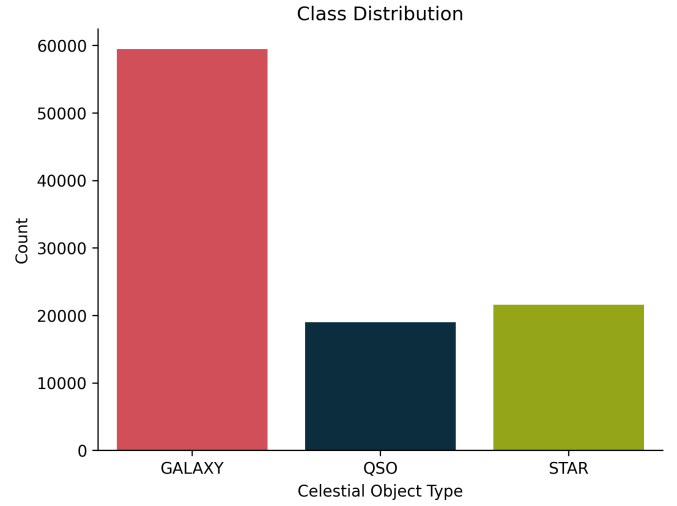


Fig. 4. Distribution of celestial object classes.

B. Evaluation Metrics

To comprehensively assess the performance of the classification models, we utilized a combination of overall and class-specific metrics. **Accuracy** was selected as the primary baseline metric to measure the general proportion of correct predictions across all classes.

However, relying solely on accuracy can be misleading in datasets with natural class imbalances. Therefore, we also generated **Confusion Matrices** for both models. These matrices are visually crucial for analyzing the exact distribution of true positives and misclassifications, allowing us to identify which specific astrophysical classes the models struggle to differentiate. Furthermore, we evaluated the **Precision, Recall, and F1-score** for each class. These metrics provide deeper insights into the models’ ability to correctly identify the minority class (QSOs) without being biased by the overwhelming presence of the majority class (Galaxies).

C. Computational Resources and Frameworks

All computational experiments, data processing, and model training were conducted using the Python 3 programming language within the Google Colaboratory (Colab) cloud environment. The implementation relied on several standard open-source libraries: **pandas** and **NumPy** were utilized for efficient tabular data manipulation and mathematical operations; **scikit-learn** (Pedregosa et al., 2011) provided the foundational framework for data preprocessing (such as the *StandardScaler* and *LabelEncoder*), dataset splitting, the implementation of the machine learning models, and the extraction of the evaluation metrics (including the classification reports and confusion matrices). Finally, **seaborn** and **matplotlib** were employed to generate the data visualizations used in our exploratory data analysis and performance evaluation.

IV. RESULTS AND DISCUSSION

A. Evaluation Results

The classification models were evaluated on the reserved testing set of 20,000 observations. Both models demonstrated exceptional predictive capabilities, successfully separating the astronomical classes, although the ensemble approach proved significantly more robust.

As summarized in Table I, the Logistic Regression baseline model achieved an overall accuracy of 95.6%. While this is a highly satisfactory result for a linear classifier, the Random Forest outperformed it, reaching an impressive accuracy of 98.1%. To fully understand these outcomes beyond general accuracy, we extracted the F1-scores for each individual class, which highlight the models’ balance between precision and recall.

TABLE I
PERFORMANCE METRICS COMPARISON

Model	Accuracy	F1 (Galaxy)	F1 (QSO)	F1 (Star)
Logistic Regression	95.6%	0.96	0.91	0.97
Random Forest	98.1%	0.98	0.95	1.00

B. Interpretation and Discussion

A deeper analysis of the classification reports and confusion matrices reveals the specific strengths and limitations of each mathematical approach, particularly regarding the minority class.

The Logistic Regression model struggled slightly with Quasars, achieving a recall score of 0.88 for this class. This indicates that the linear decision boundary misclassified 12% of the true quasars, predominantly confusing them with galaxies. This confusion aligns perfectly with astrophysical reality: quasars are hyperactive supermassive black holes residing at the centers of distant galaxies. Consequently, their photometric profiles often bleed into the spectral signatures of the host galaxies, creating a non-linear overlap that linear models struggle to resolve.

In contrast, the Random Forest classifier, using Dataset A, successfully navigated these complex boundaries. It elevated the QSO F1-score to 0.95 and achieved great performance in stellar classification, accurately identifying 100% of the stars (F1-score of 1.00). Stars are visually and physically distinct from the other two categories because they are local to our galaxy, whereas galaxies and quasars are extragalactic entities. The tree-based architecture of the Random Forest successfully isolated this fundamental difference.

C. Feature Importance Analysis

To understand the internal decision-making process of our most accurate model, we extracted the feature importances from the trained Random Forest classifier. Because this model was trained on Dataset A, it had access to all raw photometric filters and engineered color indices. However, as shown in Figure 6, the algorithm did not weigh all features equally.

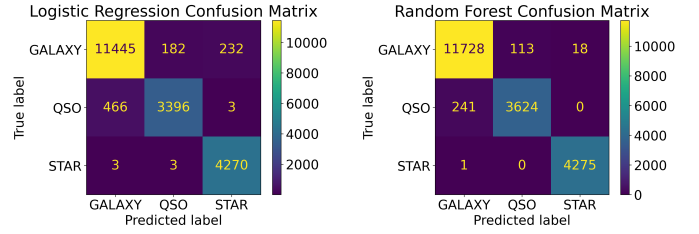


Fig. 5. Confusion Matrices for Logistic Regression (left) and Random Forest (right).

Instead, it identified *redshift* as the overwhelmingly dominant variable for classification.

This computational result strongly aligns with established astrophysical principles. Redshift measures how much an object’s light is stretched as it moves away from Earth. Because the universe is expanding, the further away an object is, the faster it is moving away from Earth, which naturally results in a higher redshift value (Carroll and Ostlie, 2017). Therefore, redshift acts as a highly reliable indicator of cosmic distance, which effectively separates our broad target categories. Stars are located nearby within our own Milky Way galaxy, so they exhibit a redshift close to zero. In contrast, standard Galaxies are typically observed at moderate cosmic distances, while Quasars—due to their extreme luminosity—can be detected much further away, dominating the highest redshift values.

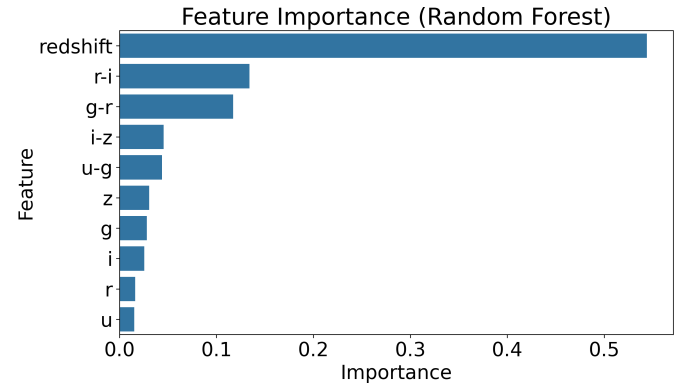


Fig. 6. Random Forest feature importance highlighting the dominance of redshift in the classification process.

V. CHALLENGES AND SOLUTIONS

During the development of this project, several technical and conceptual challenges were encountered:

- Data Leakage and Irrelevant Metadata:** *Challenge:* The raw dataset contained telescope operational data and specific identifiers (e.g., *spec_obj_ID*, *run_ID*). Retaining these could cause data leakage, allowing the models to artificially achieve high accuracy by memorizing specific observation tags rather than learning true astrophysical properties. *Solution:* These non-physical metadata columns were identified and dropped from the dataset.

- **Multicollinearity and Scale Discrepancies:** *Challenge:* The dataset exhibited multicollinearity among the photometric filters, and significant scale differences between the color measurements and *redshift*. This combination drastically destabilizes the mathematical solver of linear models like Logistic Regression. *Solution:* We branched our preprocessing pipeline to create Dataset B, dropping the highly correlated variables (g, i, z) and applying a *StandardScaler*. This ensured proper convergence and fair feature evaluation for the linear model without compromising the Random Forest, which utilized the comprehensive, unscaled Dataset A.
- **Extracting Physical Meaning:** *Challenge:* Relying solely on raw brightness magnitudes was insufficient for the models to accurately capture the subtle differences between the classes. *Solution:* Guided by astronomical principles (Abraham et al., 2011), we engineered four standard color indices ($u - g, g - r, r - i, i - z$). These differences act as direct indicator for surface temperature, significantly enhancing the models' ability to establish robust decision boundaries.
- **Spatial Generalization (α and δ):** *Challenge:* The celestial coordinates (Right Ascension and Declination) were initially considered for the models. While they could theoretically help identify local clusters of stars or galaxies, relying on them posed a major risk to generalization. A model trained on coordinates would learn to classify objects based on the specific regions of the sky mapped by the SDSS observatories, failing if applied to data from different astronomical surveys. *Solution:* We intentionally discarded both α and δ . This forced the algorithms to act as universal classifiers, predicting celestial classes based exclusively on intrinsic physical properties (photometry and redshift) rather than their arbitrary locations in the sky.

VI. CONCLUSION

In this project, we successfully developed and evaluated machine learning pipelines to automate the classification of celestial objects using photometric data from the Sloan Digital Sky Survey. By shifting away from traditional spectroscopic methods, we demonstrated that standard optical filters combined with redshift data provide a highly effective foundation for automated astrophysics. Through Exploratory Data Analysis and feature engineering, we addressed multicollinearity and extracted physical temperature profiles via color indices, transforming raw data into meaningful astrophysical representations.

Our comparative analysis between a Logistic Regression baseline and a Random Forest classifier highlighted the necessity of non-linear algorithms when dealing with complex astronomical phenomena. While the linear model achieved a highly respectable 95.6% accuracy, it struggled with the spectral overlap between Galaxies and Quasars at intermediate cosmic distances. The Random Forest model, effectively utilized both the primary distance indicator (redshift) and the

secondary thermal profiles (color indices) to resolve these edge cases.

Achieving an overall accuracy of 98.1% and a perfect stellar classification, this ensemble approach proves to be a powerful and scalable solution. Ultimately, this study confirms that machine learning is an indispensable tool capable of processing the massive influx of data expected from future astronomical surveys with exceptional speed and accuracy.

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